

Factors Influencing Students' Intentions to Use Generative AI for Learning

A Computational Human-Computer Interaction and Data Science Study

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Abstract

Generative artificial intelligence has quickly become part of how students search for information, complete schoolwork, study, and learn new concepts. However, students may differ in their willingness to use these systems based on how useful, accessible, and manageable they perceive them to be. This study examines factors associated with students' intentions to use generative AI for learning. The analysis uses survey microdata containing 336 response records ($N = 336$) and focuses on four technology-adoption constructs: Perceived Usefulness (PU), Perceived Ease of Use (PEOU), Self-Efficacy (SE), and Subjective Norms (SN). Behavioral Intention (BI) serves as the primary outcome variable. A Python-based computational workflow was developed to clean and transform the survey data, construct composite measures, evaluate descriptive statistics, and perform multivariable linear regression modeling.

The results indicated that Perceived Usefulness was the strongest positive predictor of Behavioral Intention ($B = 0.438$, $p < 0.001$), followed by Self-Efficacy ($B = 0.324$, $p < 0.001$) and Perceived Ease of Use ($B = 0.180$, $p = 0.002$). Subjective Norms were not

significantly associated with Behavioral Intention after controlling for the other predictors ($B = 0.029$, $p = 0.461$). The overall multivariable regression model was statistically significant, $F(4, 331) = 144.9$, $p < 0.001$, and explained 63.7% of the variance in Behavioral Intention ($R^2 = 0.637$, adjusted $R^2 = 0.632$). These findings suggest that perceived usefulness, self-efficacy, and perceived ease of use were more strongly associated with students' reported intentions to use generative AI than perceived social expectations. The study demonstrates how reproducible computational methods can be applied to Human-Computer Interaction research involving emerging AI technologies.

1. Introduction

Generative AI has changed how people interact with computers. Instead of relying only on traditional search engines, menus, or fixed interfaces, users can communicate with AI systems through natural language and receive responses that adapt to their questions. This change is particularly important in education. Students can use generative AI to explain difficult concepts, brainstorm ideas, summarize information, practice problems, and support independent learning. However, simply having access to an AI system does not mean that every student will choose to use it.

From a Human-Computer Interaction perspective, an important question is what makes students willing to interact with generative AI in the first place. Understanding these factors can help researchers and developers better understand how students perceive and adopt emerging AI technologies.

Technology adoption research has commonly examined factors such as perceived usefulness and perceived ease of use when studying people's willingness to use new

technologies (Davis, 1989; Venkatesh & Davis, 2000). Research on computer self-efficacy has also demonstrated the importance of users' confidence in their ability to interact with technology (Compeau & Higgins, 1995). More recent research has applied technology-acceptance frameworks to the adoption of generative AI and ChatGPT in educational settings (Albayati, 2024; Alshammari & Babu, 2025; Shahzad et al., 2024).

This study approaches the question computationally by examining the relationships between four technology-adoption constructs and students' Behavioral Intention to use generative AI for learning.

2. Research Framework and Methodology

2.1 Research Questions and Hypotheses

Primary Research Question: What factors are most strongly associated with students' intentions to use generative AI for learning?

Secondary Research Questions:

1. Is Perceived Usefulness associated with Behavioral Intention?
2. Is Perceived Ease of Use associated with Behavioral Intention?
3. Is Self-Efficacy associated with Behavioral Intention?
4. Are Subjective Norms associated with Behavioral Intention?
5. Which factors remain associated with Behavioral Intention when the predictors are analyzed simultaneously?

Hypotheses: Perceived Usefulness was hypothesized to have a positive association with Behavioral Intention and was expected to be the strongest predictor in the multivariable model.

Perceived Ease of Use, Self-Efficacy, and Subjective Norms were also expected to have positive associations with Behavioral Intention, although their strengths were expected to differ. Because this study uses observational secondary data, these hypotheses concern statistical associations rather than causal relationships.

2.2 Research Variables and Dataset

This study uses an existing secondary survey dataset containing 336 response records ($N = 336$). The survey measures use a 7-point Likert scale ranging from 1 (Strongly Disagree) to 7 (Strongly Agree).

The primary constructs selected for analysis were:

- Perceived Usefulness (PU): Independent variable, measured using four survey items (PU1–PU4).
- Perceived Ease of Use (PEOU): Independent variable, measured using four survey items (PEOU1–PEOU4).
- Self-Efficacy (SE): Independent variable, measured using three survey items (SE1–SE3).
- Subjective Norms (SN): Independent variable, measured using three survey items (SN1–SN3).
- Behavioral Intention (BI): Dependent outcome variable, measured using four survey items (BI1–BI4).

The dataset also contains additional constructs that were not included in the primary regression model because the research question focused specifically on PU, PEOU, SE, and SN.

2.3 Computational Methods and Data Processing

Python served as the primary computational environment. Pandas and NumPy were used for data processing and transformation. SciPy and Statsmodels were used for statistical analysis and regression modeling, while Matplotlib and Seaborn were used for visualization.

The computational workflow consisted of the following stages:

Existing Dataset → Data Ingestion → Data Validation → Data Cleaning → Composite Score Construction → Descriptive Analysis → OLS Regression → Model Diagnostics → Visualization → HCI Interpretation

The raw dataset was inspected for data types, missing values, invalid responses, duplicate records, unexpected values, and extreme observations. Data-processing decisions were documented to support reproducibility.

2.4 Composite Variable Construction

Each primary construct was measured using multiple survey items. Composite scores were calculated by taking the arithmetic mean of the corresponding items for each participant.

Perceived Usefulness: $PU = (PU1 + PU2 + PU3 + PU4) / 4$

Perceived Ease of Use: $PEOU = (PEOU1 + PEOU2 + PEOU3 + PEOU4) / 4$

Self-Efficacy: $SE = (SE1 + SE2 + SE3) / 3$

Subjective Norms: $SN = (SN1 + SN2 + SN3) / 3$

Behavioral Intention: $BI = (BI1 + BI2 + BI3 + BI4) / 4$

Using composite scores allowed related survey items to be represented as a single measure for each construct. The same scoring procedure was applied consistently across the dataset.

3. Statistical Analysis

3.1 Descriptive Statistics

Descriptive statistics were calculated for the five primary construct scores. These included the mean (M), standard deviation (SD), minimum, maximum, and sample size.

3.2 Multiple Linear Regression

An Ordinary Least Squares (OLS) multivariable linear regression model was used to examine the simultaneous associations between Perceived Usefulness, Perceived Ease of Use, Self-Efficacy, Subjective Norms, and Behavioral Intention.

The regression model was: $BI_i = \beta_0 + \beta_1(PU_i) + \beta_2(PEOU_i) + \beta_3(SE_i) + \beta_4(SN_i) + \epsilon_i$

where:

- BI_i represents the Behavioral Intention score for participant i.
- β_0 represents the intercept.
- β_1 represents the estimated association between Perceived Usefulness and Behavioral Intention while holding the other predictors constant.
- β_2 represents the estimated association between Perceived Ease of Use and Behavioral Intention while holding the other predictors constant.

- β_3 represents the estimated association between Self-Efficacy and Behavioral Intention while holding the other predictors constant.
- β_4 represents the estimated association between Subjective Norms and Behavioral Intention while holding the other predictors constant.
- ε_i represents the residual error for participant i .
- The regression analysis reported unstandardized coefficients (B), standard errors, t-statistics, p-values, 95% confidence intervals, R^2 , and adjusted R^2 .
- The coefficients were interpreted as statistical associations rather than causal effects.

3.3 Statistical Significance and Effect Size

The primary significance level was $\alpha = 0.05$.

Statistical significance was not treated as the only measure of importance. The analysis also considered regression coefficients, confidence intervals, R^2 , and adjusted R^2 to evaluate the strength and explanatory value of the observed relationships.

4. Results

4.1 Descriptive Statistics

Multi-item Likert indicators were aggregated into composite scale scores for each participant ($N = 336$). Perceived Ease of Use and Behavioral Intention recorded the highest average scores, both with $M = 5.48$, whereas Subjective Norms exhibited the lowest mean score ($M = 4.77$).

Table 1. Descriptive Statistics for Primary Construct Scores

Construct Scale	Abbreviation	Items	Mean (M)	Std. Dev. (SD)	Min	Max
Perceived Usefulness	PU	4	5.43	1.25	1	7
Perceived Ease of Use	PEOU	4	5.48	1.19	1	7
Self-Efficacy	SE	3	5.22	1.29	1	7
Subjective Norms	SN	3	4.77	1.46	1	7
Behavioral Intention	BI	4	5.48	1.34	1	7

Overall, the descriptive results indicate that participants reported relatively positive scores across the primary constructs. Perceived Ease of Use and Behavioral Intention had the highest average scores, while Subjective Norms had the lowest.

4.2 Multivariable OLS Regression Analysis

The specified OLS model explained 63.7% of the variance in Behavioral Intention ($R^2 = 0.637$, adjusted $R^2 = 0.632$). The overall regression model was statistically significant, $F(4, 331) = 144.9$, $p < 0.001$.

Table 2. Multivariable OLS Regression Results Predicting Behavioral Intention

Predictor	Unstandardized Coefficient (B)	Std. Error	t	p	95% Confidence Interval
Intercept	0.2926	0.227	1.289	0.198	[-0.154, 0.739]
Perceived Usefulness (PU)	0.4376	0.054	8.173	< 0.001	[0.332, 0.543]
Perceived Ease of Use (PEOU)	0.1796	0.057	3.126	0.002	[0.067, 0.293]
Self-Efficacy (SE)	0.3235	0.055	5.922	< 0.001	[0.216, 0.431]
Subjective Norms (SN)	0.0287	0.039	0.737	0.461	[-0.048, 0.105]

Table Statistics:

- Perceived Usefulness had the largest positive coefficient among the four predictors (B = 0.4376, SE = 0.054, t = 8.173, p < 0.001, 95% CI [0.332, 0.543]).
- Self-Efficacy also showed a statistically significant positive association with Behavioral Intention (B = 0.3235, SE = 0.055, t = 5.922, p < 0.001, 95% CI [0.216, 0.431]).
- Perceived Ease of Use showed a smaller but statistically significant positive association with Behavioral Intention (B = 0.1796, SE = 0.057, t = 3.126, p = 0.002, 95% CI [0.067, 0.293]).
- Subjective Norms had a small positive coefficient, but the association was not statistically significant after controlling for the other predictors (B = 0.0287, SE = 0.039, t = 0.737, p = 0.461, 95% CI [-0.048, 0.105]).
- The intercept was not statistically significant (B = 0.2926, p = 0.198).

4.3 Summary of Predictor Results

Perceived Usefulness was the strongest positive predictor in the model. Holding the other predictors constant, a one-unit increase in Perceived Usefulness was associated with an estimated 0.44-unit increase in Behavioral Intention.

Self-Efficacy was the second strongest positive predictor. A one-unit increase in Self-Efficacy was associated with an estimated 0.32-unit increase in Behavioral Intention while holding the other predictors constant.

Perceived Ease of Use was also positively associated with Behavioral Intention. A one-unit increase in Perceived Ease of Use was associated with an estimated 0.18-unit increase in Behavioral Intention after controlling for the other predictors.

Subjective Norms did not show a statistically significant association with Behavioral Intention in the multivariable model. After accounting for Perceived Usefulness, Perceived Ease of Use, and Self-Efficacy, Subjective Norms did not explain substantial additional variation in Behavioral Intention in this dataset.

5. Discussion

5.1 HCI and Technology Adoption Implications

The findings indicate that Perceived Usefulness was the strongest positive predictor of students' reported intentions to use generative AI for learning. This supports the primary hypothesis that students who perceive generative AI as more useful tend to report stronger intentions to use it.

Self-Efficacy was the second strongest positive predictor. Students reporting greater confidence in their ability to interact with technology also tended to report stronger intentions to use generative AI. From a Human-Computer Interaction perspective, this

suggests that adoption may depend not only on the capabilities of the technology but also on users' confidence in navigating and using it.

Perceived Ease of Use also demonstrated a statistically significant positive association with Behavioral Intention, although its coefficient was smaller than those of Perceived Usefulness and Self-Efficacy. This suggests that usability remains relevant to students' intentions to use generative AI, while usefulness and self-efficacy showed stronger associations in this dataset.

Subjective Norms produced a different result from the original hypothesis. Although a positive association was expected, Subjective Norms were not statistically significant after the other predictors were included in the model. This suggests that perceived social expectations did not provide substantial additional explanatory value once functional and individual factors were taken into account.

Taken together, the findings suggest that students' intentions to use generative AI for learning may be more closely associated with their perceptions of the technology's usefulness and their confidence in using it than with perceived social expectations.

5.2 Model Diagnostics

Regression diagnostics were used to evaluate the appropriateness of the OLS model and identify potential violations of model assumptions.

Multicollinearity: The model condition number was 55.1. Predictor Variance Inflation Factors (VIF) remained below the commonly used threshold of 5.0, indicating no substantial evidence of problematic multicollinearity among the predictors.

Autocorrelation: The Durbin-Watson statistic was 1.747, suggesting no substantial evidence of first-order autocorrelation.

Residual Normality: The model produced an Omnibus statistic of 28.999 ($p < 0.001$), with skewness of -0.507 and kurtosis of 4.667 . These results indicate some departure from normal residuality. Because the data are based on bounded Likert-scale measures, some deviation from normality is possible. The results should therefore be interpreted with the limitations of the data structure in mind.

Influential Observations: Cook's distance values remained below 0.5 across the observations, indicating no observations with exceptionally large influence on the fitted model based on this diagnostic.

These diagnostics provide evidence regarding the behavior of the fitted model but do not establish that all OLS assumptions are perfectly satisfied.

6. Limitations and Ethical Considerations

6.1 Observational Design

This study identifies statistical associations rather than causal relationships. Although several predictors were statistically significant, the regression model cannot demonstrate that changes in those variables directly cause changes in Behavioral Intention.

6.2 Self-Reported Data

The analysis relies on self-reported survey responses. Participants may interpret survey questions differently or use rating scales differently, introducing potential response or measurement bias.

6.3 Original Study Design

The researcher did not control the original survey design, participant recruitment, or data-collection process. Consequently, limitations in the original study may affect the present analysis.

6.4 Generalizability

The findings should not automatically be generalized to all students. The extent to which the results generalize depends on the population and sampling methodology of the original dataset.

6.5 Measurement

The analysis depends on the reliability and validity of the original survey measures. Composite scores assume that the selected items adequately represent their respective constructs.

6.6 Secondary-Data Ethics

This project analyzes an existing dataset rather than collecting new participant responses. No direct participant recruitment or interaction was conducted as part of this analysis. The study is intended to use the dataset in accordance with its applicable terms of use, protect any restricted information, report results in aggregate, and avoid disclosure of personally identifying information.

7. Conclusion

This study examined factors associated with students' intentions to use generative AI for learning using a computational Human-Computer Interaction and data science approach.

Using survey data from 336 response records, the analysis evaluated Perceived Usefulness, Perceived Ease of Use, Self-Efficacy, and Subjective Norms as predictors of Behavioral Intention. The multivariable OLS model explained 63.7% of the variance in Behavioral Intention and was statistically significant overall.

Perceived Usefulness was the strongest positive predictor, followed by Self-Efficacy and Perceived Ease of Use. Subjective Norms were not statistically significant when the four predictors were considered simultaneously.

These findings suggest that students' reported intentions to use generative AI may be particularly associated with whether they perceive the technology as useful and whether they feel confident using it. Ease of use also appears to play a meaningful role, while perceived social expectations provided less additional explanatory value in this model.

Because the study uses observational secondary data, these relationships should not be interpreted as causal effects. Nevertheless, the findings demonstrate how computational methods, statistical modeling, and Human-Computer Interaction principles can be combined to investigate how people choose to interact with emerging AI technologies.

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Data and Code Availability

The computational workflow, statistical analysis, and visualization code are documented in the project's GitHub repository:

<https://github.com/AashigaM/ai-hci-research>

The repository contains the Python-based analysis workflow and supporting documentation used to process the survey data, construct composite variables, perform statistical modeling, and generate visualizations. No confidential participant-level information is intended to be included in the public repository.